

JONY KARMAKAR

Data Science Graduate | SQL, Python, AI Expertise | Digital Solutions & Innovation

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Ås, Norway

EXPERIENCE

Research Collaborator

Norges Miljø- og Biovitenskapelige Universitet

- 02/2026 - Present | Ås, Norway
- Participated in the project "Advancing Controlled Environment Agriculture Using AI," funded by the Research Council of Norway.
 - Contributed to research on vision-based plant growth estimation, supporting ongoing scientific work in the project.
 - Built and evaluated computer vision workflows for growth signals, including dataset preparation, labeling support, and reproducible experimentation.
 - Tested different temporal deep learning models such as MAML for analyzing the growth of plants.

KS Analyst (Voluntary)

Samfunnet i Ås

- 10/2023 - 12/2025 | Ås, Norway
- Collected and consolidated operational and financial data from multiple sources to support decision support for 5+ major events and 10+ weekly operations.
 - Designed and maintained KPI reports tracking 50+ metrics, enabling resource optimization, variance analysis, and improved decision-making.
 - Performed forecasting and driver analysis on time-dependent data, translating findings into decision-ready insights and recommended actions.

Research and Development Intern

Unipolar Automation Technologies

- 11/2021 - 07/2022 | Dhaka, Bangladesh
- Supported the development of a subscription-based Wi-Fi system for public spaces, serving 500+ users daily and improving system reliability through optimized backend processes.
 - Collaborated across engineering teams to define technical requirements, optimize data architecture, and align system performance with business objectives.
 - Contributed to operational improvement by vetting technical requests, preparing quotations, and ensuring clear communication between internal teams.

EDUCATION

Masters in Data Science

Norges Miljø- og Biovitenskapelige Universitet

- 08/2022 - 05/2025 | Ås, Norway
- Relevant courses: Data-driven decision making, Machine learning, Deep learning, Data management and analysis, advanced programming in Python, and Image processing.
 - Thesis: "AI-Based Foul Detection and Card-booking in Soccer" - Designed a deep learning pipeline using CNN-based and Transformer-based models for multi-view video analysis in sports e.g. soccer.

Average Grade
B / Avg

B.Sc. in Computer Science and Engineering

Bangladesh Army International University of Science & Technology

- 03/2016 - 08/2020 | Cumilla, Bangladesh
- Relevant Courses: Data Structure and Algorithm, Object Oriented Programming (Python), Database Management System (SQL), Data Mining, and System Analysis & Design
 - Thesis: "Implementation of a Novel Hybrid Model for Bengali Text Summarization using Conventional Extractive Method and Text-Rank Algorithm"

GPA
3.97 / 4.0

PROJECTS

Operational Analytics Dashboard for Fish Health & Weather Data

- https://github.com/JonyKarmakar/StreamsyncProject
- Created an interactive analytics dashboard using Python, Spark, Cassandra, and Streamsync to visualize trends and forecast fish health and weather data. Implemented data cleaning, integration flows, and ML models for forecasting.
- Technical Highlights:** Python, Docker, Spark, Cassandra, API, Data Quality, Data Collection & Integration, Data Visualization, Predictive Modeling, Dashboarding, Deployment. (VS Code, Streamsync)

SUMMARY

Data Science graduate with hands-on experience building practical AI solutions, analytics workflows, and decision-support dashboards. Strong in Python and SQL, with applied experience in deep learning and computer vision projects that required end-to-end pipelines, reproducible experimentation, and clear documentation. Comfortable translating technical needs into usable prototypes and collaborating across disciplines to deliver solutions that are scalable and reliable in real-world settings.

SKILLS

Programming & Analytical Tools

Python (Pandas, NumPy, Scikit-learn), SQL, Jupyter Notebooks, Power BI (DAX, Data Modeling), Excel (Automation & Analysis)

Data Analytics & Reporting

Exploratory Data Analysis (EDA), Statistical Visualization, Time-Series Analysis, KPI Reporting, Trend & Anomaly Detection, Forecasting

Business Intelligence & Decision Support

Dashboard Development, KPI Analytics, DAX Measures, Drill-Down & Interactive Reports, Excel Analytics, Data Storytelling

Machine Learning & Statistical Analysis

Predictive Modeling, Regression Analysis, Anomaly Detection, Hypothesis Testing, Time-Series Forecasting, Feature Engineering, Model Evaluation & Validation, Evaluation Metrics

Data Management & Quality

Data Extraction, Cleaning & Transformation, Data Validation & Quality Checks, Structured Datasets Maintenance, Reproducible Workflows

Data Governance & Privacy

Data Quality, GDPR, Access Control, Aggregation

Cloud & Infrastructure

Azure, GCP, Docker, REST APIs, CI/CD, Git

Collaboration & Project Tools

Agile Methods, Jira, Trello, GitHub, Google Workspace, Cross-Functional Coordination

Core Competencies

Adaptability, Critical Thinking, Communication, Teamwork, Creativity, Cross-Functional Collaboration, Initiative, Continuous Learning, Structured Problem-Solving

TRAINING / COURSES

Networking and Basic Server Management

BAIUST ICT WING

PROJECTS

Sales Dashboard & Reporting (Power BI & Google Sheet)

Developed an interactive sales dashboard using Power BI to analyze Samfunnet's sales data. Integrating data from various sources and used DAX for advanced calculations along KPI tracking.

- Technical Highlights:* Data Management, Data Modeling, DAX, Reporting, KPIs, Dashboard, Role-Based Access Control (RBAC), Forecasting (PowerBI), Spreadsheet Analytics.

Deep Learning Pipeline Development

<https://github.com/JonyKarmakar/DeepLearning>

Developed Deep Learning pipeline for image classification, recognition, segmentation using U-Net, and VAE models.

- Technical Highlights:* Python, CNN, U-Net, Variational Autoencoder, Keras, Model Evaluation (VS Code).

Data Analysis & Workflow Automation with Classical ML Technique

<https://github.com/JonyKarmakar/MachineLearning>

Trained and tested machine learning models with feature engineering, cross-validation, hyperparameter tuning, and model evaluation for improved accuracy.

- Technical Highlights:* Python, data preprocessing, model optimization, predictive modeling (VS Code).

Tailwind Traders Sales, Profit & Executive Dashboard (Power BI)

The project was designed as a scenario-based approach simulating the demands of a real business environment, requiring proficiency in the techniques using Power BI.

- Technical Highlights:* Role-Based Access Control (RBAC), Data Visualization Software, SQL, Data Modeling, Dashboard, Power BI, Excel, Data Analysis, Data Management, Data Security

Spotify Data Analysis

<https://github.com/JonyKarmakar/MachineLearning/tree/main/CA1>

Analyzed Spotify user data for patterns in popularity, audio features, and genre traits. Used distribution, correlation, regression, and clustering to explore track feature's impact on listener engagement.

- Technical Highlights:* P Python, pandas, EDA, Correlation Analysis, Regression, Clustering, Statistical Testing (VS Code)

3D animation of a Rail crossing using OpenGL

<https://github.com/JonyKarmakar/RailCrossingAnimation>

Developed a 3D animation using OpenGL Library with OOP concepts in C++.

- Technical Highlights:* C++, animation, OOP, graphical modeling (CodeBlocks).

Modeling the Ecosystem in Rossumøya

<https://github.com/JonyKarmakar/BioSim>

Project fully focusing on use of OOP and logics using python. A set of tests and efficiency has been checked for the whole project.

- Technical Highlights:* Python (OOP), ecosystem simulation, data modeling (VS Code).

Text Summarization using TextRank Algorithm

<https://github.com/JonyKarmakar/TextRank>

Implemented a text summarization tool using the TextRank algorithm to extract meaningful summaries from large bodies of Bengali text. This project required processing large text datasets, applying NLP techniques, and optimizing the summarization model to ensure high accuracy and efficiency.

- Technical Highlights:* Python, NLP, Data cleaning, machine learning, TextRank algorithm (IntelliJ).

Sentiment Analysis of customer comments for products from an e-commerce site

<http://github.com/JonyKarmakar/SentimentAnalysis>

Experimented sentiment analysis on e-commerce customer reviews written in Bengali, applying NLP and machine learning techniques to classify sentiments.

- Technical Highlights:* Python, Bayesian Theorem, Machine Learning, Natural Language Processing (NLP), NLTK, data preprocessing (Google Colab).

TRAINING / COURSES

Competitive Programming Boot Camp

Bangladesh Advanced Computing Society (BACS)

CERTIFICATIONS

Microsoft Power BI Data Analyst Professional Certificate

Microsoft, Coursera

Google IT Automation with Python, Professional Certificate

Google, Coursera

Microsoft Certified: Azure Fundamentals (Ongoing)

Microsoft

KEY ACHIEVEMENTS

 **Budget Analysis Expertise**
Being part of managing a NOK 1.5M budget, leading to a 10% cost reduction through financial analysis.

 **KPI Dashboard Development**
Developed dashboards tracking 50+ metrics, improving decision-making efficiency by 25%.

 **Research Collaborating with SIMULA**
Collaborated with SIMULA for Master's Thesis and was part of FROZIFY team where we worked on implementation of AI in soccer analytics.

AWARDS

 **Vice Chancellor's Academic Achievement Award**
BAIUST

 **Runner-up in Project Showcasing (Software)**
BAIUST Computer Club

 **Dean's Scholarship for Academic Excellence**
BAIUST

LANGUAGES

English
Proficient 

Norwegian
Intermediate 

Bengali
Native 